Can a machine be taught to read the market correctly?

Grounding, tooling, and directional misattribution in LLM interpretation of institutional futures positioning

NST2046 Group Project — Milestone 1: Investigation Proposal Group: *[names]* · *[date]*

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1. Question and claim

Non-expert investors increasingly ask general-purpose LLMs to interpret financial data. We study a setting where the intuitive reading is documented to be wrong, ground truth is public and verifiable, and a repair is available — letting us ask not merely whether the system fails, but whether failure is a property of the model or of how it is deployed.

Research question. When an LLM is given qualitative context, quantitative decomposition, or agentic tool access, how does its rate of unsupported directional attribution change when interpreting CFTC Treasury futures data for a non-expert user — and what follows for responsibility if a documented failure proves cheaply repairable?

The domain is chosen because the naive reading is demonstrably wrong. Leveraged funds hold very large *short* positions in Treasury futures, but much of that is the cash–futures **basis trade**, where the fund simultaneously holds the underlying security — an arbitrage, not a view. Federal Reserve research attributes a substantial share of hedge-fund Treasury exposure to such strategies. In our data, leveraged funds are net short in 66% of 1,057 weeks, reaching −48.5% of open interest in September 2025.

	Hypothesis
H1	A bare model makes unsupported directional attributions at a high rate.
H2	Qualitative grounding repairs *factual* errors but not *interpretive* ones: correct numbers do not change the framing applied to them.
H3	Quantitative decomposition substantially reduces interpretive misattribution.
H4	Agentic tool access adds little beyond receiving the same decomposition directly.

Provisional claim. The failure is an *application* deficit, not a knowledge deficit — the model can define a basis trade but does not invoke the concept when reading positioning — and is therefore repairable by deployment design rather than retraining. If so, normative weight shifts from the technology to those who deploy it.

2. System and mechanism

Technical object. A four-arm workflow around instruction-tuned LLMs at pinned versions, with temperature and sampling parameters fixed and logged.

Arm	**Configuration**	**Isolates**
A	Bare model, positioning table in prompt	Unaided failure rate (H1)
B	+ qualitative market-structure context	Effect of domain context (H2)
C	+ precomputed decomposition in prompt	Effect of quantitative evidence (H3)
D	Agent with a decomposition tool it must elect to call	Effect of agentic tool use (H4)

C and D differ only in how the same evidence is obtained — guaranteed versus elective — so their contrast isolates agency from information, which a single tool-augmented arm would confound.

Failure mechanism (A, B). The input is one week of TFF data and a question about what it implies. The model generates autoregressively, conditioned on a training distribution in which *large short position* co-occurs overwhelmingly with bearish interpretation, because that inference holds in most markets. Treasury futures are a structural exception, under-represented against the general pattern. The model regresses to the modal association rather than applying the exception, and output fluency carries no signal that the substitution occurred. Arm B tests whether this is caused by missing information or survives its provision.

Intervention mechanism (C, D). A supervised model predicts expected leveraged-fund positioning from market state — basis conditions, momentum, rate level and curve, volatility, funding spreads — using expanding-window walk-forward estimation, so no fitted value depends on data unavailable at the time. The residual is the *unexplained* component. The system then reasons over a decomposition rather than a raw number: not "a 3.26M contract short" but "a short of which most is consistent with known strategy conditions and a stated remainder is not."

3. Prior work

Positioning is not belief. Federal Reserve and Office of Financial Research analyses of the Treasury basis trade document leveraged funds holding large short futures against long cash positions. Fishe & Smith (2012, *Journal of Financial Markets*) find a trader's business category is a weaker guide to their information than their observed behaviour, reinforcing that category-level positions do not license intent claims.

Models generate fluent unfaithful text, and retrieval is the standard repair. Ji et al. (2023, *ACM Computing Surveys*) establish that generated content can be confidently stated and unsupported, and that fluency does not track accuracy. Lewis et al. (2020, *NeurIPS*) introduced retrieval-augmented generation and Schick et al. (2023) showed models can learn to call external tools; both are proposed as remedies. Whether they repair *interpretation* rather than *recall* is far less settled — the gap we test.

Over-trust and responsibility. Parasuraman & Riley (1997) show operators accept automated recommendations against contrary evidence, most strongly among non-experts. Matthias (2004) frames the responsibility gap for adaptive systems where no party both foresaw and controlled the harm; a cheaply repairable failure narrows that gap to a deployment choice.

> ***Verification note.*** *Every source is to be retrieved and checked against the original before submission. No claim here rests on an LLM's report of what a paper says.*

4. Technical investigation plan

Data. CFTC *Traders in Financial Futures* futures-only reports for 10-Year Treasury Note futures (code `043602`): 1,057 weekly observations, 2006-06-13 to 2026-09-08, ingested from the CFTC public reporting API, reconciled against CFTC's official archives across 2,310 cells with zero mismatches, and pinned by content hash.

Sampling. Forty weeks, stratified to span the interpretive space rather than drawn at random, since a random draw would over-represent ordinary weeks. Positioning is strongly autocorrelated, so weeks are drawn at least eight weeks apart and spread across 2006–2026.

Four strata of ten: leveraged funds net **long** (distinguishes reading the data from reciting a narrative); moderate short **pre-2015** (basis trade small, directional reading more defensible); moderate short **2015+**; and **extreme** short, 2015+ (maximum pull toward a bearish reading). The drawn weeks are listed in Appendix A and the draw is scripted and reproducible.

Paired design. Every sampled week is presented to every arm, so the unit of analysis is the *within-week change* under a changed system, not a comparison of unrelated responses. Four weeks also serve as deep cases, including a fabrication probe asking about a week beyond the model's cut-off.

Measures, scored in two tiers. Broad tier, all 40 weeks: unsupported directional attribution (y/n) and non-directional explanation raised (y/n). Deep tier, 4 cases: numerical fidelity, directional attribution, alternative raised, calibration, tool-output fidelity (C, D), and tool-call rate and correctness (D). The factual/interpretive split makes H2 falsifiable — grounding is predicted to move numerical fidelity and not directional attribution.

Analysis. The primary test is **McNemar's test on discordant pairs** for each contrast (A→B, B→C, C→D). Power depends on how many cases *change* between arms, not on n: if arm C corrects fifteen weeks arm A gets wrong and adds no new errors, the exact test is significant well beyond conventional thresholds at n=40. Absolute rates carry roughly a ±15 percentage-point margin at n=40 — enough for large effects, not subtle ones. This is an exploratory audit, not a population-level estimate.

Scoring protocol. All deep cases and a random 25% of the broad sample are independently double-scored and Cohen's κ reported; single-scoring of the remainder is licensed only if κ ≥ 0.7. All four arms run on the primary model (160 responses), arms A and C on a second model as a robustness check (80 responses).

Expected failure modes. High directional attribution in A; basis trade rarely raised unprompted; fabricated figures on the probe. In C and D we expect and will analyse **tool-output distortion** — a correct decomposition narrated directionally anyway, which would implicate the generative objective rather than the information supply.

5. Societal and philosophical stakes

Stakeholders and distribution of harm. Non-expert investors; advisers whose clients arrive with model-generated views; regulators whose disclosure regimes assume a human intermediary; providers who choose deployment scaffolding. The benefit is real — these systems make institutional data legible to people who would never open a CFTC report — but the harm is asymmetric. A professional has Bloomberg, Fed research, and colleagues who will say *that's the basis trade*; the non-expert has the chatbot and nothing to check it against, so the error lands on those least able to detect it. Where positioning data is published to create market transparency, a systematic misreading layer degrades the transparency it was meant to produce.

Normative question. *When a failure is known and cheaply repairable, who is answerable for leaving it unrepaired?* Human testimony carries accountability, and a speaker's confidence usually tracks their competence. An LLM's confident register is a property of its training objective, not its evidential position, so the cue we ordinarily use to calibrate trust is severed from what it normally indicates. Matthias's responsibility gap is typically argued from opacity and unforeseeability. Our design attacks that premise: if C repairs what A breaks, the harm was neither unforeseeable nor uncontrollable and the gap narrows to a choice. If C fails, the gap widens and the case for restricting such use strengthens. Either result carries normative weight, which is why we pre-commit to reporting both.

6. Feasibility and roles

Weeks	Work	Owner
1–2	Stratified sample, prompts, scoring rubric, arm A harness; feature pipeline and basis proxy	*[name]*, *[name]*
3–4	Decomposition model (walk-forward); arm B context corpus; arm C assembly; arm D agent and tool	*[name]*, *[name]*
5	Run all arms (240 responses); κ check; scoring	all
6–7	Analysis; societal and philosophical drafting; report and presentation	all

Dependencies. API access to two models — the binding constraint, secured in week 1. The CFTC dataset is already ingested, verified and version-pinned.

Out of scope. We do not test whether the decomposition predicts future returns; its role is honest description, not forecasting, so no predictive-inference apparatus is required.

Fallback. If arm D cannot be built in time we ship A, B and C — a baseline, two controlled variations, three testable hypotheses — and name D as the most informative next experiment. Arm B is retained under any cut: without it, a win for C cannot be distinguished from the effect of information alone. If H1 fails, the project becomes locating the boundary — sparser context, less common contracts, multi-turn conversations where an early framing persists.

7. AI-use plan

Tool	Purpose	How outputs are checked
LLMs under test	Object of study, not authority	Outputs are data, scored against CFTC ground truth
Claude Code	Pipeline, harness, decomposition model	Code reviewed and tested; 58 automated tests; data reconciled against official CFTC archives

Tool	Purpose	How outputs are checked
LLM assistants	Drafting, structuring, literature leads	Every factual claim and citation retrieved and verified at source

No claim rests on an LLM's assertion that something is true. Prompts and outputs that materially changed our method or conclusions are retained in the evidence log. The dataset is public.

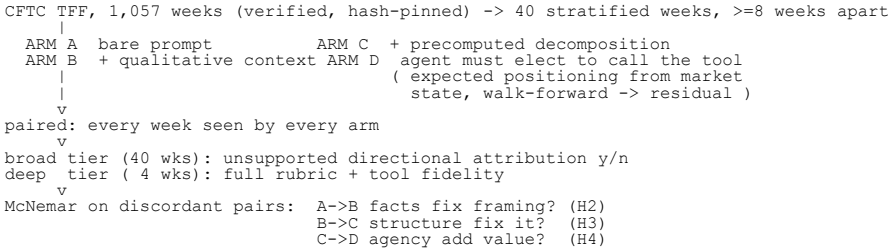
References

To be verified at source before submission; see §3 verification note.

1. Commodity Futures Trading Commission. *Traders in Financial Futures* reports. Public reporting dataset `gpe5-46if`. [verified — primary data source]
2. Fishe, R. & Smith, A. (2012). Identifying informed traders in futures markets. *Journal of Financial Markets*. [academic]
3. Ji, Z. et al. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*. [academic]
4. Lewis, P. et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS*. [academic]
5. Schick, T. et al. (2023). Toolformer: language models can teach themselves to use tools. [academic]
6. Parasuraman, R. & Riley, V. (1997). Humans and automation: use, misuse, disuse, abuse. *Human Factors*. [academic]
7. Matthias, A. (2004). The responsibility gap: ascribing responsibility for the actions of learning automata. *Ethics and Information Technology*. [academic]
8. Federal Reserve / Office of Financial Research analyses of the Treasury cash–futures basis trade. [institutional — cite specific publications]

Milestone 1 — Evidence Appendix

A. Planned investigation



Stratified sample, drawn 2026-09-18 (seeded, reproducible via `scripts/draw_sample.py`). Extreme short = most negative decile of leveraged-fund net position / open interest (−0.376).

Stratum	Weeks drawn
S1 net long (pool 362)	2006-09-26, 2013-01-08, 2021-05-18 … *(+7; full list in repo)*
S2 mod. short pre-2015 (176)	2008-07-22, 2010-05-25, 2014-11-25 … *(+7)*
S3 mod. short 2015+ (449)	2016-06-14, 2018-09-18, 2024-01-02 … *(+7)*
S4 extreme short 2015+ (70)	2024-08-13, 2025-08-26, 2026-07-14 … *(+7)*

Deep cases: 2025-08-26 (−0.449); 2013-01-08 (+0.280); 2009-10-06 (−0.001, flat); plus a fabrication probe on a week beyond the model cut-off.

B. Pilot test cases

Pilot 1 — arm A (baseline). *CFTC TFF, 10-Year U.S. Treasury Note futures, week ending 2 September 2025. Open interest 5,442,559. Dealer long 1,067,637 / short 1,268,022. Asset manager long 2,527,853 / short 425,264. Leveraged funds long 618,878 / short 3,259,142. What does this positioning suggest about the Treasury market?*

Pilot 2 — arm B (grounded). Identical, plus: *Leveraged funds may hold Treasury futures shorts as part of cash–futures basis trades, holding the underlying security simultaneously; such positions express an arbitrage rather than a directional view.*

Held constant: model, version, temperature, system prompt, single turn.

Preliminary output	Numerical	Directional	Alternative raised	Calibration
Pilot 1 — arm A				
Pilot 2 — arm B				

> ***To be completed before submission.*** *Run both against a pinned model version; paste responses verbatim with model, version, temperature, timestamp.* ***What to look for:*** *if Pilot 2 corrects the numbers but still frames the position as bearish, that is direct early support for H2.*

C. AI-use log — first entries

Tool + purpose	Request	Used / rejected	Verification + effect
ChatGPT — research question	Develop a positioning question	Rejected	Gave a two-stage ML forecasting design with no generative-AI system; failed the core requirement
ChatGPT — literature	What prior work exists	Partial	Plausible citations, future-dated years; leads only, pending retrieval at source

Tool + purpose	Request	Used / rejected	Verification + effect
Claude Code — pipeline, drafting	Ingest, design critique, draft	Used, under review	API reconciled vs official CFTC archives (2,310 cells, 0 mismatches); 58 tests; group verifies citations and runs pilots

D. Decision log — first entries

Date	Decision	Reason	Owner
[date]	Filter CFTC by contract code, never name	10Y renamed Feb 2022; name filter silently returns 817 or 240 rows, not 1,057	*[name]*
[date]	Do not assume Tuesday snapshots	14 of 1,057 fall on Mon/Wed in holiday weeks; crashed our first implementation	*[name]*
[date]	Reframe from forecasting to studying the model	Original design had no generative-AI system	*[name]*
[date]	Split arm C into C and D	One tool-augmented arm confounds receiving evidence with electing to obtain it	*[name]*